

Elicitation of Consumption and Survey Design

Session 3

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The five components of consumption expenditures

Or: "Nondurable consumption and services".

Consumption expenditures in the LSMS surveys generally fall into one of five categories, and are typically elicited in different modules.

1. **Food** — purchased, own-produced, and received as gifts
2. **Non-food, non-durable** — fuel, transport, clothing, school fees, health
3. **Durables** — *value of services*, or "use value", not purchase price
4. **Housing** — rent, actual or imputed
5. **Other Services** — Not really measured at all

List Length

Beegle et al (2012): the *number* of goods asked has a large effect on recall, and on that evidence many NSOs have lengthened the list. Just for food:

Survey	Earlier rounds	Later rounds
Ethiopia	25 → 26 → 55	73 → 74
GhanaLSS	58 → 58 → 117 → 112	114 → 114 → 179 → 400
Burkina Faso	55 → 130	152
Uganda	61 → 63 → 70 → 71	77 → 77 → 117 → 116
Nigeria	90 → 90 → 96 → 104	116 → 116 → 110 → 110

- Should make measured total consumption more accurate;
- BUT: Really complicates comparisons across time.
- And: Can greatly increase cost of fielding the survey.

Timing & Recall

The model treats time as discrete. What in the field corresponds to a "period"? LSMS practice reveals what the designers thought.

Food Usually a seven-day recall, with interesting variation and some explicit experiments (Beegle et al. 2012) on how well respondents remember purchases.

Non-food expenditures or "semi-durables" Think things like clothing, inventories of kerosene.... Often a 30-day recall.

Durables Bicycles, radios, fans,... Often a year-long recall for purchases, and/or an inventory of the *stock* of durables.

Asking about food: GhanaLSS 2016-17

At each of six visits, five days apart. *Maize, rice, cassava, garden eggs, smoked fish, cooking oil, ...* — 179 items:

Food purchases

*"How much was spent and what quantity was acquired of ... since my **last visit**?"*

amount, quantity, unit, per visit; a diary kept between visits.

Autoconsumption

*"How much of own produced ... was consumed by the household since my **last visit**?"*

Asking about food: Uganda 2019-20

The case of Uganda is more standard than Ghana; there's a standard food expenditure module which is basically the same across the eight LSMS-ISA countries. For Uganda there are 129 food items in 2019-20.

Asking about food: Uganda's single visit

Single Visit

*"How many [UNITS] of [ITEM] did your household consume out of [SOURCE] at home **during the last 7 days?**"*

Then [SOURCE] takes values:

- "purchases"
- "purchases away from home"
- "out of home production"
- "received in-kind"

Asking about non-food: the window stretches

Ghana 2016-17

Asked once, at the last visit:

How much was spent on...in the past 12 months altogether?"

Frequently-bought non-food (toothpaste, haircuts, transport) also inside the six-visit food module instead.

Asking about non-food: Uganda 2019-20

Uganda 2019-20

Asked once:

"How much did the household spend in total on these [UNITS] of [ITEM] during the last [PERIOD]?"

PERIOD:

- 30 days for non-durables (e.g., soap, electricity, tires, stamps);
- 365 days for semi-durables (e.g., clothing, furniture, spoons, school fees).

Asking about durables: the stock, not the purchase

Attempts to assess the value of stocks, rather than flows.

Ghana 2016-17

*"Does any member of the household own [ASSET]? How long ago was it obtained? What was its purchase price? **How much could you sell it now** in Ghana cedis?"*

Uganda 2019-20

*"Does any member of your household own [ASSET] at present? How many? What is the **total estimated value** of the [ASSET] owned at present?"*

Asking about housing: rent, actual and imagined

Ghana 2016-17

*"How much does the household pay **in cash** towards the rent?"*

An amount plus a time unit the *respondent* picks (daily ... yearly). Owner-occupiers skip the block; no imputed rent is ever asked.

Asking about housing: Uganda 2019-20

Uganda 2019-20

Rent is part of the same 30-day recall on non-food expenditures:

"How much did the household spend in total on these [RENT OF RENTED HOUSE] during the last 30 days?"

Part of the same module as non-durables; same question format.

"What would be the total value of [IMPUTED RENT OF OWNED HOUSE]... if you were to sell at the farm gate, during the last 30 days?"

Food prices: which price?

Food *expenditures* are quantities times prices. Which prices?

- farmgate price received — what the household gave up
- local market price — what it would have cost
- these differ by the marketing margin, said to be 30–50%

Deaton & Zaidi: value at the price the household *faces*. In practice, a median unit value by item, cluster and round, falling back up the hierarchy when cells are thin.

Which of the two does a survey's own valuation question measure?

Food prices: what GLSS7 asks, and what it gets

Section 8 Part H, question 9: **"for how much would you sell one unit of ... now?"** A farmgate question, by its wording (though hypothetical).

Against what neighbours *paid*, same item, unit and cluster:

- Ghana 2016-17, 322 cells: median ratio **1.000**; a third tie exactly
- Uganda, eight waves, the same question: **1.000** pooled
- Uganda also records sales: the valuation is **1.23** times the sale price; buyers paid **1.60** times it

Asked what they would sell a bowl of maize for, a household answers with the price of a bowl of maize. The margin is real — 1.3 to 2.2 where a survey records sales.

Food quantities, and what was never bought

Much of the food in an LSMS household was never bought:

- Purchased at market
- Consumed out of own production
- Gift/Transfer
- Consumed outside the household (?)

Not a canard. In Ghana 2016-17 own production is **28.7%** of the value of food; 12,191 of 13,942 households record some. Deaton and Zaidi's Table 3.1 implies 31.9%.

But it reaches your aggregate only if you *value* it: from 1991-92 on, the survey records own production in quantities, not cedis.

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GLSS7's answer: a diary, visited six times

The most recent GhanaLSS is ambitious about frequent purchases: a diary, and an enumerator visiting six times at five-day intervals.

The form (reading/): six visit columns headed *2nd* to *7th*, asking **"...since my last visit?"**, and no price column anywhere.

Expensive; two experiments have asked whether it is worth it.

Scott & Amenuvegbe (1990) Ghana, and the reason GLSS asks this way

Beegle et al. (2012) Tanzania, eight modules, randomly assigned

The diary

CLUSTER NUMBER:

HOUSEHOLD NUMBER:

NAME OF HOUSEHOLD HEAD: DATE:

A R DAY MTH Y E

INSTRUCTION: RECORD ALL PAYMENTS MADE AND ESTIMATE ALL OWN PRODUCED GOODS CONSUMED.

Date Amount	Place of Purchase/ Own Produce	Description of Item Purchased or	Consumed	Qty	Unit	Price

"Record all payments made and *estimate all own produced goods consumed*": the diary asks for a price on own produce. Section 9B, filled from it, does not.

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How fast does recall decay?

Scott & Amenuvegbe (1990), 13 frequently purchased items in Ghana:

- Reported daily expenditure falls steadily as the recall window lengthens from one day to one week, then flattens between one and two weeks
- Roughly **2.9% lost for every day added**, after reweighting away purchase cycles and start-up bias
- The decline is larger for items bought more often
- A one-year recall is simply erratic

So a weekly recall understates frequent purchases by around **20%** — and that was, in their words, "an abnormally well controlled field situation."

What actually drives the error

Beegle et al. (2012): eight modules randomly assigned across ~4,000 Tanzanian households, benchmarked against a personal diary.

- Recall *length* is the small knob: 7 vs 14 days moves food spending **7%**
- Item *list* design is the big one: **21%** between 58 items and 11 categories
- "Usual month" fails and contaminates the next module
- Household diaries miss about **19%** of what personal diaries catch

Poverty ranges from **47.5% to 66.8%**, by module alone.

Timing: what is a period?

For food, a day, probably: people schedule around meals. Surveys pick something longer, and each choice buys one error with another.

- A *diary* of what was eaten, meal by meal, is the ideal and the burden: hard to sustain, and paying for it buys a record, not an accurate one
- A *short* window is cheap but noisy: day-to-day variation in intake is not the error we are after
- *Usual* consumption asks the respondent to do our averaging for us, and to share our idea of "typical"; Beegle et al. found it fails outright

Inventories: purchases are not consumption

Storable staples — dry maize, rice, oil — are bought in bulk and drawn down. Perishables — fruit, fresh fish — are bought as eaten.

- A short window catches the fruit and misses the maize sack, or catches the sack and triples the week
- Averages out across households if purchase timing is unrelated to the visit; does not average out *within* one, which is where the line is applied
- Who buys in bulk is a matter of wealth: cash on hand, a dry store, a refrigerator. So the noise is on the rich, and on the staples

Measuring consumption despite inventories

Two approaches:

- Repeated visits over thirty days, as in GLSS7.
- Ask what was *consumed*, as Section 8H does, rather than only what was purchased, as in Section 9B.

Both address the mismatch between the timing of purchases and consumption.

Price Indices

Two different price indices, meant to address two different problems:

- **Temporal:** prices rise between rounds. Use the CPI, or better, a survey-based Paasche index built from the survey's own unit values.
- **Spatial:** prices differ across regions *within* a round, often by more than they differ across a decade. A national poverty line applied to undeflated nominal consumption will find poverty wherever prices are high.

A household-specific Paasche index

Deaton & Zaidi recommend a household-specific Paasche index

$$P_i = \left[\sum_j s_{ij} \left(\frac{p_{ij}}{p_j^0} \right)^{-1} \right]^{-1},$$

with s_{ij} household i 's budget share on item j .

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Durables and housing

Durables: what enters consumption is the *flow of services*, not the purchase. For a stock S with service life L and interest rate r ,

$$\text{use value} \approx S \left(r + \frac{1}{L} \right).$$

Housing: for owner-occupiers there is no rent, so impute it.

- hedonic regression of log rent on housing characteristics, among renters
- predict for owners
- fragile where the rental market is thin, which is most rural areas

Intra-Household Allocation

Adult equivalence

A household of six is not six times as well off as a household of one with the same total consumption. Two adjustments, often conflated:

- **Household composition:** children cost less than adults —
 $\Rightarrow$ adult-equivalent scale $A_i = \sum_k a_{\text{age}, \text{sex}}$
- **Economies of scale:** shared goods, so cost rises less than proportionately — $\Rightarrow n_i^\theta$, with $\theta \in (0, 1)$

Combined: $c_i = \frac{C_i}{A_i^\theta}$.

θ is not estimated in this session; it is *assumed*. Report the poverty rate for several values of it.

Barten: children re-price goods

A^θ makes a child a fraction of an adult *for everything*. Barten (1964): composition changes *which* goods a household needs,

$$\max_c U\left(\frac{c_1}{m_1(z)}, \dots, \frac{c_J}{m_J(z)}\right) \quad \text{s.t.} \quad \sum_j p_j c_j = x,$$

so a household with demographics z is the reference household at prices $p_j m_j(z)$. A child makes milk dear; substitution does the rest.

Barten: what the scale depends on

- the scale is a ratio of cost functions, $e(u, p \circ m(z))/e(u, p)$: it depends on prices, and in general on u
- A^θ is the case where every m_j is the same number: children change the *size* of the bundle, never its *shape*
- a shape that depends on z is not consistent with homotheticity

Constructing Aggregate Welfare Measures

Poverty measures

The Foster–Greer–Thorbecke class: for poverty line z ,

$$P_{\alpha} = \frac{1}{N} \sum_{i:c_i < z} \left(\frac{z - c_i}{z} \right)^{\alpha}.$$

- $\alpha = 0$: headcount ratio. Ignores *how* poor the poor are.
- $\alpha = 1$: poverty gap. The transfer needed, per capita, as a share of z .
- $\alpha = 2$: squared gap. Weights the poorest most; the only member of the family that satisfies the transfer axiom.

Using sample weights: $P_{\alpha} = \sum_i w_i \left(\frac{z - c_i}{z} \right)^{\alpha} / \sum_i w_i$.

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- **Lorenz curve:** cumulative share of consumption against cumulative share of population. Everything else is a summary of it.
- **Gini:** twice the area between the Lorenz curve and the 45-degree line. Bounded, familiar, and insensitive where it matters most.
- **Generalized entropy** $GE(\alpha)$: decomposable into within- and between-group parts. $GE(0)$ is mean log deviation, $GE(1)$ is Theil.

Inequality: Atkinson

The opening slide's $I = U(\int w dF) - \int U(w) dF$ is in utils, which nobody reports. Let w_{ede} solve $U(w_{\text{ede}}) = \int U dF$: the level which, given to everyone, yields the same total utility as the actual spread. Then $I = U(\bar{w}) - U(w_{\text{ede}})$, and

$$A = 1 - \frac{w_{\text{ede}}}{\bar{w}}$$

is the same judgement in cedis per cedi, with $U(w) = w^{1-\varepsilon}/(1-\varepsilon)$ and ε assumed, like θ .

$A = 0.3$: this society would give up 30% of mean consumption to have the rest shared equally. Prefer the Lorenz curve to any scalar.

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One index, or several?

Everything here divides nominal expenditure by a scalar: a Paasche index over space, a CPI or PPP over time, an equivalence scale over composition. Three scalars, one household, one number out.

- one deflator is exact at **rank one**: budget shares do not vary with total expenditure
- but they do — Engel's law, which opened this session
- a line *and* a deflator are exact at **rank two** (Gorman), which Engel's law permits
- so on what grounds is a two-index procedure adequate?

Hold the question: next session answers it for Uganda.

Exercises

1. The poverty line used above is fake. Build a real one: take the food bundle consumed by households in the second and third deciles, price it at national median unit values, and scale it to 2,900 kcal per adult equivalent.
Recompute P_0, P_1, P_2 . Notebook: `poverty_line.ipynb`
2. Deflate spatially. Construct a regional Paasche index from the survey's own unit values, apply it, and report how much of the north-south poverty gradient survives.
Notebook: `spatial_deflation.ipynb`
3. Vary θ in $c_i = C_i/A_i^\theta$ over $[0.5, 1.0]$ and plot the headcount ratio against it. Over what range does the *ranking* of regions change? Notebook: `equivalence_scales.ipynb`